

ADRIAN VILCHEZ

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EDUCATION

University of California, Santa Cruz
B.S. Computer Science

Exp. June 2026

TECHNICAL STRENGTHS

Programming Languages	Python, C/C++, Haskell
Tools	AWS, Docker, Firebase, Git
Concepts	Backend Programming, LLM/ML, QC
Libraries	Qiskit, OpenCV, PyTorch, Numba
Languages	Fluent in English and Spanish

RELEVANT COURSEWORK

Quantum Computing	Chaos Theory	Functional Programming
Computer Systems and Concurrency	Proofing for Computer Science	Computer Architecture
Embedded OS	Algorithm Analysis	Data Structures and Algorithms

PROJECTS

Quantum Computing Capstone Fall 2025 - Winter 2026
Understanding Implementation of Shor's Algorithm in Quantum Computing

- Explained the fundamentals of Quantum Technology to advanced concepts such as QFT, QPE, and Shor's Algorithm

Nano Interpreter and Type Checker - Functional Programming Project Winter 2026
Haskell

- Designed Interpreter with features up to standardized error identification and type polymorphism

SafeWatch - WebScraping Project Winter 2026
TS, React

- Created and Implemented live alerting application for localized ICE Raids

Custom Neural Network Library Spring - Summer 2025
C/C++

- Designed a fully modular neural network framework with custom and advanced ADTS for memory-efficient representation.
- Implemented standard features such as forward propagation with sigmoid activation, MSE cost function, and several gradient update functions

WORK EXPERIENCE

ACE Parking
Valet Parking Attendant

Summer/ Winter. 2023, Summer 2024
Palo Alto, CA

- Delivered professional, high-quality guest services while coordinating efficiently with team members.

P.A.L. (Police Athletic League)

Youth Mentor

Sept. 2021 - June 2022

Redwood City, CA

- Led workshops and activities fostering teamwork, skill development, and community engagement.
- communication, patience, and leadership

LEADERSHIP AND INVOLVMENT

CruzHacks, Association for Computing Machinery (ACM), Society of Hispanic Professional Engineers (SHPE)

- Active member in mentorship and professional development communities at UCSC.